How Much Knowledge Can You Pack Into the Parameters of a Language Model?

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Abstract

It has recently been observed that neural language models trained on unstructured text can implicitly store and retrieve knowledge using natural language queries. In this short paper, we measure the practical utility of this approach by fine-tuning pre-trained models to answer questions *without access to any external context or knowledge*. We show that this approach scales with model size and performs competitively with open-domain systems that explicitly retrieve answers from an external knowledge source when answering questions. To facilitate reproducibility and future work, we release our code and trained models.[1]

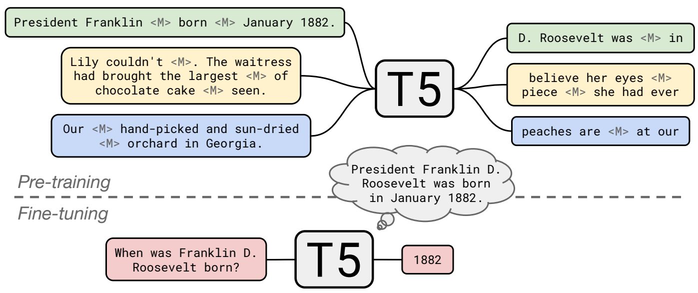

Figure 1: T5 is pre-trained to fill in dropped-out spans of text (denoted by <M>) from documents in a large, unstructured text corpus. We fine-tune T5 to answer questions without inputting any additional information or context. This forces T5 to answer questions based on "knowledge" that it internalized during pre-training.

1 Introduction

Big, deep neural language models that have been pre-trained on unlabeled text have proven to be extremely performant when fine-tuned on downstream Natural Language Processing (NLP) tasks (Devlin et al., 2018; Yang et al., 2019; Liu et al., 2019; Lan et al., 2019; Raffel et al., 2019). Interestingly, it has also recently been observed that these models can internalize a sort of implicit "knowledge base" after pre-training (Petroni et al., 2019; Jiang et al., 2019; Talmor et al., 2019). This behavior is potentially useful because 1) the knowledge is built up by pre-training on unstructured and unlabeled text data, which is freely available in huge quantities on the Internet (Raffel et al., 2019; Wenzek et al., 2019), and 2) it is possible to retrieve information using informal natural language queries since these pre-trained language models excel when fine-tuned on natural language understanding tasks.

Past work investigating "language models as knowledge bases" has typically tried to understand the scope of the information stored in the model using synthetic tasks that are similar to the pre-training objective (Petroni et al., 2019; Jiang et al., 2019) and/or measure reasoning capabilities (Talmor et al., 2019). In this work, we take a different approach by evaluating the capability of language models on the practical task of open-domain question answering – specifically, we fine-tune the model to answer questions *without access to any external knowledge or context*. To do so, the model must parse a natural language query and "look up information" stored in its parameters.

Most past work on question answering either explicitly feeds pertinent information to the model alongside the question (for example, an article that contains the answer (Rajpurkar et al., 2016; Zhang et al., 2018; Khashabi et al., 2018; Clark et al., 2019)) or allows the model to retrieve information from an external knowledge source (Berant et al., 2013; Chen et al., 2017). By feeding the model the input question alone, we can determine how much knowledge it has stored in its param-

* Equal contribution. Noam suggested trying T5 on open-domain QA and coded and ran initial experiments on TriviaQA showing improved performance with model size. Adam wrote the code and ran most experiments. Colin set the research scope, wrote the paper, and ran a few experiments.

[1] https://goo.gle/t5-cbqa

eters while measuring its performance on a useful real-world problem. We refer to this task as "closed-book question answering".

A separate question we address in this work is whether models with more parameters end up storing more information. It has been shown that transfer learning performance on many downstream tasks tends to improve as the model size and amount of unsupervised pre-training increases (Radford et al., 2019; Liu et al., 2019; Raffel et al., 2019). In this work, we leverage the pre-trained "T5" models released by Raffel et al. (2019), the largest of which has around 11 billion parameters. By measuring knowledge retrieval capabilities on models of various sizes, including models that have an order of magnitude more parameters than considered in past work, we can explore how well our approach scales.

2 Background

Question Answering The task of training a model to either select or output the correct answer to a given question is referred to as "question answering". The most popular variant of this task feeds the model some "context" containing the answer (for example, a paragraph from an encyclopedia article) alongside the question (Rajpurkar et al., 2016; Zhang et al., 2018; Khashabi et al., 2018; Clark et al., 2019). Models can be trained either to indicate the span of the context that contains the answer or output the text of the answer itself. Since this format can be seen as reading some text and answering a question about it, it has been referred to as "reading comprehension".

A more difficult variant is "open-domain question answering" (Prager, 2006), where the model can be asked arbitrary context-independent questions (e.g. well-known facts or historical details). It is typically assumed that the model can access an external collection of knowledge when answering questions (e.g. a structured knowledge base or unstructured text corpus), but the model is not given any information about where in the collection the answer appears. The reading comprehension task can be considered a simplified version of open-domain question answering where the model is provided with the oracle context to answer a given question. As an analogy, the open-domain question answering system acts as if it is taking an **open-book** exam where it can find and use information in an external source of knowledge.[2]

In this work, we consider open-domain question answering with the additional constraint that the model is *not* allowed to access any external knowledge whatsoever when answering questions. Instead, the model itself must be pre-trained to store knowledge in its parameters before being fine-tuned to answer questions. In one view, this can be seen as an alternative way to approach open-domain question answering where instead of learning to access external knowledge the model needs to have "memorized" it in order to answer questions; in another view, this constraint creates a third and potentially more ambitious variant of the question answering task. A model that answers questions in this way is metaphorically similar to a student taking a **closed-book** exam, where the student must study and memorize all pertinent information before taking the test.

Transfer Learning with Language Models In the past few years, it has become increasingly common to pre-train a language model using an unsupervised objective on a large, unstructured text corpus before fine-tuning it on a downstream task of interest (Dai and Le, 2015; Howard and Ruder, 2018; Radford et al., 2018). The popularity of this form of "transfer learning" is attributable to its empirical success on many NLP tasks (Peters et al., 2018; Devlin et al., 2018; Yang et al., 2019; Lan et al., 2019; Raffel et al., 2019). Loosely speaking, the pre-training step may provide the model with some generally-useful awareness of meaning, syntax, and "world knowledge". In question answering in particular, most state-of-the-art systems use some form of transfer learning.

Currently, the most popular model architectures used in transfer learning for NLP are Transformer-based (Vaswani et al., 2017) "encoder-only" models like BERT (Devlin et al., 2018). These models can produce a single prediction for each input token and have been applied to reading comprehension-style question answering by predicting which tokens of the context contain the answer. Encoder-only models are not applicable to closed-book question answering because no context is provided to extract the answer span from. An alternative to encoder-only models, recently advocated by Raffel et al. (2019), is to treat ev-

[2]While our definition of open-book is the same as in the OpenBookQA dataset introduced by Mihaylov et al. (2018), we do not directly address multi-hop inference in this work.

ery NLP task as a text-to-text problem using an encoder-decoder Transformer. When this framework is applied to question answering, the model is trained to generate the literal text of the answer in a free-form fashion. Despite the potential difficulty of generating rather than extracting the answer, Raffel et al. (2019) demonstrated state-of-the-art results on the SQuAD (Rajpurkar et al., 2016), MultiRC (Khashabi et al., 2018), BoolQ (Clark et al., 2019), and ReCoRD (Zhang et al., 2018) reading comprehension tasks.

The text-to-text framework is directly applicable to closed-book question answering since the model can be trained to generate an answer with or without any additional information in its input. Crucially, fine-tuning a text-to-text model to answer questions without any context requires that the model retrieve information from its parameters that it learned during pre-training. Radford et al. (2019) considered a similar task to evaluate the zero-shot question answering capabilities of a language model. The concurrent "RELIC" and "EAE" models of Ling et al. (2020) and Févry et al. (2020) learn representations for an explicitly predefined set of entities and are evaluated on the same closed-book variant of TriviaQA that we consider. Relatedly, Petroni et al. (2019) show that it is possible to manually convert some questions to a fill-in-the-blank format amenable to an encoder-only model (e.g. "Who developed the theory of relativity?" gets mapped to "The theory of relativity was developed by ____").

3 Experiments

Datasets We consider the following open-domain question answering datasets: *Natural Questions* (Kwiatkowski et al., 2019), a dataset of questions from web queries, each accompanied by a Wikipedia article containing the answer; *WebQuestions* (Berant et al., 2013), comprising questions from web queries matched to corresponding entries in FreeBase (Bollacker et al., 2008); and *TriviaQA* (Joshi et al., 2017), a collection of questions from quiz league websites where each question is accompanied by pages from web and Wikipedia searches that may contain the answer. In this work, we only make use of the questions from each dataset – *we completely ignore the matching documents supplied for each question*.

For WebQuestions and TriviaQA we follow the standard evaluation procedures where each predicted answer is compared to the ground-truth after both are lowercased and stripped of articles, punctuation, and duplicate whitespace (Rajpurkar et al., 2016). For Natural Questions, we evaluate using both 1) the standard "open-domain" version as used e.g. by (Lee et al., 2019; Min et al., 2019b,a; Asai et al., 2019) where the model is only required to produce a single normalized answer and 2) the standard multi-answer variant used with reading comprehension systems (Kwiatkowski et al., 2019). We review the details of Natural Questions evaluation in appendix A.

Note that Natural Questions and TriviaQA have private test sets, so standard practice on their open-domain variants is to report performance on the development sets. However, we also include our results on the official TriviaQA test set by fine-tuning on the unfiltered training set and submitting our test set predictions to the leaderboard for the Wikipedia domain. We urge future work to adopt this approach to help ensure the validity of results and avoid potentially overfitting to a public set.

Training We leverage the pre-trained models provided by Raffel et al. (2019), referred to as the "Text-to-Text Transfer Transformer" (T5). The original T5 models were pre-trained on a multi-task mixture including an unsupervised "span corruption" task on the C4 dataset as well as supervised translation, summarization, classification, and reading comprehension tasks. Note that none of the reading comprehension datasets used for pre-training T5 overlap with the question answering datasets that we consider in this paper. In order to measure how performance scales with model size, we perform experiments with the Base (220 million parameters), Large (770 million), 3B (3 billion), and 11B (11 billion) variants of T5. Given that the T5 models were pre-trained on a multitask mixture including question answering, we also report performance using the "T5.1.1" checkpoints, which were pre-trained on unlabeled data only.[3]

For fine-tuning the T5 checkpoints, we follow the procedure used in Raffel et al. (2019) without any additional hyperparameter tuning: We use the AdaFactor optimizer (Shazeer and Stern, 2018) with a constant learning rate of 0.001, 10% dropout rate, and a batch size of 196,608 tokens. We halve the batch and double the dropout rate for WebQuestions due to its small size. For the T5.1.1 checkpoints, we follow the same procedure

[3] `https://goo.gle/t5-checkpoints`

but with a dropout rate of 5% for all three datasets.

For evaluation, we follow the procedure used in Lee et al. (2019): for each dataset, we hold out 10% of the training set as a validation split, fine-tune a model from the remaining 90% of examples, and select the best-performing checkpoint for final evaluation on the test set. While we chose to train for 20,000 steps, our validation accuracy typically plateaued after only a few hundred steps and showed no signs of overfitting.

We decode the model's predictions by choosing the most likely token at each timestep. To map question answering tasks to the text-to-text format, we simply feed the question with a task-specific prefix into the model as input and train it to predict the literal answer text as output.

Salient Span Masking Recently, Guu et al. (2020) found that a "salient span masking" (SSM) pre-training objective produced substantially better results in open-domain question answering. This approach first uses BERT (Devlin et al., 2018) to mine sentences that contain salient spans (named entities and dates) from Wikipedia. The question answering model is then pre-trained to reconstruct masked-out spans from these sentences, which Guu et al. (2020) hypothesize helps the model "focus on problems that require world knowledge". We experimented with using the same SSM data and objective to continue pre-training the T5 checkpoints for 100,000 additional steps before fine-tuning for question answering.

Results Our results on the open-domain Natural Questions, WebQuestions, and TriviaQA tasks are shown in table 1. Notably, performance on each dataset improves as the model size increases, with either T5-11B or the comparably-sized T5.1.1-XXL (pre-trained only on unlabeled data) performing best in every case. Further, we find that using Guu et al. (2020)'s SSM pre-training produces a substantial boost in performance. T5.1.1-XXL with SSM ultimately achieves state-of-the-art on WebQuestions and our largest models beat most other methods on Natural Questions and TriviaQA. Importantly, all previous methods except Ling et al. (2020) and Févry et al. (2020) operate in the "open-book" setting by explicitly retrieving and using information from an external knowledge source. While our largest models are computationally intensive, we note that most open-domain question answering systems must first do an expensive lookup step over the entire knowledge corpus and then attend to a long document to extract an answer. Our approach omits both of these steps, which ultimately saves a large amount of computation and memory.

Table 1: Scores achieved by fine-tuning T5 on the open-domain Natural Questions (NQ), WebQuestions (WQ), and TriviaQA (TQA) tasks.

	NQ	WQ	TQA dev	TQA test
Chen et al. (2017)	–	20.7	–	–
Lee et al. (2019)	33.3	36.4	47.1	–
Min et al. (2019a)	28.1	–	50.9	–
Min et al. (2019b)	31.8	31.6	55.4	–
Asai et al. (2019)	32.6	–	–	–
Ling et al. (2020)	–	–	35.7	–
Guu et al. (2020)	40.4	40.7	–	–
Févry et al. (2020)	–	–	43.2	53.4
Karpukhin et al. (2020)	**41.5**	42.4	**57.9**	–
T5-Base	25.9	27.9	23.8	29.1
T5-Large	28.5	30.6	28.7	35.9
T5-3B	30.4	33.6	35.1	43.4
T5-11B	32.6	37.2	42.3	50.1
T5-11B + SSM	34.8	40.8	51.0	60.5
T5.1.1-Base	25.7	28.2	24.2	30.6
T5.1.1-Large	27.3	29.5	28.5	37.2
T5.1.1-XL	29.5	32.4	36.0	45.1
T5.1.1-XXL	32.8	35.6	42.9	52.5
T5.1.1-XXL + SSM	35.2	**42.8**	51.9	**61.6**

Having established that our approach is competitive on open-domain question answering, we now evaluate it on the standard (and more difficult) multi-answer variant of Natural Questions. Virtually all models used on this task are reading comprehension systems that select the correct answer from an oracle context. After fine-tuning, T5-11B + SSM achieves a recall of 36.2 on the validation set, which lags behind the state-of-the-art score of 51.9 from Pan et al. (2019)[4] but outperforms the best baseline published alongside the dataset (recall of 33.2 (Kwiatkowski et al., 2019)). This shows that T5 can effectively answer questions with multiple answers. We discuss additional experiments and negative results in appendix B.

Human Evaluation The benchmarks we used and the "exact match" score assume that the model directly extracts answers from an external knowledge source. In contrast, our model generates answers in a free-form fashion. We hypothesize that this results in many false negatives when an-

[4]Validation set recall scores from Pan et al. (2019) were reported in private correspondence with the authors.